Exact Transfer of Bounded Linear Recovery
and Relative Weight Hierarchies

Tavis Rudd

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Abstract

Let $I \leq \mathbb{F}_q^E$, let $P \subseteq E$ be a target set, and put $J = E \setminus P$. Shortening and puncturing $I^\perp$ onto J give $K_P \subseteq D_P \leq \mathbb{F}_q^J$, whose relative generalized Hamming weight $M_t(D_P, K_P)$ is the minimum helper union recovering t independent target combinations. For a fixed nonzero target-message subspace, we express the exact least nonconfined helper union as an optimization of prescribed-coset support costs over the complete outer functional dual, without an outer-distance gate. For an L-linear outer code $O \leq L^N$, $N \geq 2$, the condition $d(O^\perp) > r + 1$ reduces the exact criterion to

$$r < M_t(D_P, K_P) + d(I^\perp).$$

Below either exact threshold, zero-extension preserves normalized equations and supports. The one-coordinate specialization recovers the weighted-coset formula and can transfer beyond the ordinary outer support-distance gate. Positive-density and bounded service-rate transfer follow, while equal relative-weight hierarchies need not determine bounded repair reliability. ERGO-Comp implements the constructive formulas as an exact hierarchical optimizer and capacity-aware scheduler with retained witnesses; on the recorded structured instances it is 2.5–377 times faster than direct CP-SAT.

1 Introduction

Locality asks how many helpers recover one erased coordinate. Cooperative locality and (r, δ)-locality ask for recovery after several erasures, and availability asks for disjoint choices of helpers. Generalized Hamming weights give global lower bounds for cooperative recovery [22, 10, 20, 1]. Recovery sets and coordinate-indexed recovery structures record more local information [18], but support data alone discard the dual coefficients that implement the recovery equations.

This paper applies relative generalized Hamming weights to a canonical nested pair obtained by shortening and puncturing the inner dual. Relative generalized Hamming weights and the relative dimension/length profile were introduced for nested codes by Luo, Mitrpant, Vinck, and Chen, who interpreted them as access thresholds in a wiretap model [16]; later work develops their ramp-secret-sharing and algorithmic roles [13, 9, 23, 8]. In particular, Kurihara, Uyematsu, and Matsumoto characterize exact information and reconstruction thresholds of linear secret-sharing schemes by RDLPs and RGHWs. Our recovery-cost identity is this standard nested-code support invariant specialized to the puncturing–shortening pair induced by a target/helper split. The paper's principal result begins with concatenation. For every recovered target subspace, the exact finite obstruction is an optimization of joint coset-support costs over the complete outer functional dual. The ordinary outer-dual-distance gate removes all nonzero-functional terms, leaving the relative weight plus one additive inner-dual obstruction. At one target coordinate the same formula becomes the pointed weighted coset-leader theorem.

Let q be a prime power. Let $I \le \mathbb{F}_q^E$ have a full-row-rank generator matrix $G = (G_P \mid G_J)$, where P is the target set and $J = E \setminus P$. Put

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

and define the associated nested code pair

$$K_P = \ker G_J \subseteq D_P = \{a \in \mathbb{F}_q^J : G_J a \in U_P\}.$$

Equivalently, and independently of the chosen row basis of G,

$$D_P = \operatorname{punct}_J(I^\perp), \qquad K_P = \operatorname{short}_J(I^\perp).$$

We assume $0 < \dim I < |E|$, so both the represented message space and $I^\perp$ are nonzero. We use the convention $d(0) = \infty$ for the zero code. The quotient $D_P/K_P \cong W_P$ is exactly the space of target linear combinations recoverable from the helper coordinates. Here t is the dimension of a recovered subspace of W_P, not the number of erased coordinate positions. Recovering all values on P is the special case $U_P \subseteq \operatorname{im} G_J$ and $t = \dim U_P$; dependent target columns may make this dimension smaller than $|P|$.

Theorem 1.1 (Main theorem: exact finite transfer and its RGHW specialization)**.** *Let $\ell = \dim W_P \ge 1$ and $1 \le t \le \ell$. The minimum cardinality of the union of helpers in a system recovering a t-dimensional subspace of W_P is*

$$\mu_t(P) = M_t(D_P, K_P).$$

Let I be used as the represented inner code in a concatenation with an L-linear outer code $O \le L^N$ with $N \ge 2$, where L is the message space of I, and fix a block onto which O projects nontrivially. The quantity $\Gamma_{j,t}(O, I)$ in (12), obtained by minimizing the joint support of compatible inner coset representatives over the complete outer functional dual, is exactly the first nonconfined helper cost in recovered dimension t. Hence confinement through radius r holds exactly when $r < \Gamma_{j,t}(O, I)$. Moreover,

$$d(O^\perp) \ge M_t(D_P, K_P) + d(I^\perp) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^\perp).$$

If in addition $d(O^\perp) > r + 1$, this exact criterion reduces to

$$r < M_t(D_P, K_P) + d(I^\perp).$$

When the inequality holds, zero-extension gives a bijection on the normalized recovery-equation systems and their exact helper supports. Consequently, for any outer family with dual distance tending to infinity, the same statement holds for all sufficiently long concatenations at every fixed r.

The first assertion is the recovery-language specialization of the standard nested-code invariant: a recovery system identifies with a subspace $A \le D_P$ disjoint from K_P. Passing between this form and the standard definition of $M_t(D_P, K_P)$ uses a complement to $A \cap K_P$. The finite transfer formula then separates the concatenated dual by its outer functional tuple and minimizes the joint support of all compatible inner representatives. Its zero-functional sector is the least inner recovery system together with a minimum inner-dual perturbation in another block. The outer-distance gate excludes every other sector at radius r; dual-distance growth makes that gate automatic at fixed radius. For one target coordinate, the general joint support minimum is the scalar weighted coset formula of Section 4.

The same bijection yields the later transfer results. It preserves every block-local statistic determined by normalized recovery systems of cost at most r. After forgetting coefficients, it also preserves statistics determined by inclusion-minimal helper supports, including bounded repair reliability and the bounded service-rate region. This statement is block-local: the upward-closed recovery-set family in the full concatenated coordinate set also contains irrelevant cross-block supersets.

The theorem contains the previous one-coordinate criterion. If $P = \{x\}$ and $t = 1$, then $M_1(D_P, K_P)$ is the least number of helpers in a dual word through x, and

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp).$$

Thus $r < M_1 + d(I^\perp)$ is exactly $r + 1 < z_x(I)$.

The separation results delimit the information carried by this hierarchy:

$$\text{relative-weight hierarchy} \not\Longrightarrow \text{bounded repair reliability}, \qquad (K_P, D_P) \not\Longrightarrow \text{confinement threshold}.$$

The first failure comes from the overlap pattern of all recovery equations. The second compares different ambient inner-dual realizations of the same abstract nested pair; it is not a change of generator-row basis for one code.

Concatenation has long been used to construct LRCs with prescribed parameter properties [15, 12]. Jin and Fu fix a binary $[3, 2, 2]$ inner code and choose $\mathbb{F}_4$ outer codes to obtain parameter-optimal binary locality-two codes. Here the inner code and target split are arbitrary, and the question is the exact obstruction to preserving every bounded dual recovery equation. The finite formula is rank-stratified and ungated; the RGHW formula is its outer-distance specialization. We make no minimum-bandwidth claim.

The formulas are also algorithms. ERGO-Comp evaluates the labelled prescribed-coset recursion through a finite tower, retains the intermediate functionals needed for composition, and returns exact recovery witnesses and confinement thresholds. Its capacity scheduler uses the same compiled recovery choices to optimize simultaneous repairs under heterogeneous helper loads. Section 5 gives correctness and complexity bounds and compares both direct and equivalently preprocessed CP-SAT baselines.

2 Recovery sets and normalized equations

Let $C \leq \mathbb{F}_q^E$ be a linear code. We first separate four layers that are often harmlessly conflated when only locality is at issue.

Fix a target $x \in E$ and a helper bound r. A dual word $w \in C^\perp$ with $w_x \neq 0$ gives

$$c_x = \sum_{y \neq x} -\frac{w_y}{w_x} c_y \qquad (c \in C).$$

Its exact helper support is $\mathrm{supp}(w) \setminus \{x\}$. Define

$$\mathcal{H}_x^{(\leq r)}(C) = \{\mathrm{supp}(w) \setminus \{x\} : w \in C^\perp, w_x \neq 0, \mathrm{wt}(w) \leq r + 1\}.$$

The usual bounded recovery-set family is the upward closure of $\mathcal{H}_x^{(\leq r)}(C)$ inside the subsets of $E \setminus \{x\}$ of size at most r. Its inclusion-minimal members form the minimal recovery-set clutter.

Thus exact supports, recovery sets, and minimal recovery sets are distinct but determine the same monotone success event.

The coefficient layer is

$$\mathcal{P}_x^{(\leq r)}(C) = \{w \in C^{\perp} : w_x = 1, \mathrm{wt}(w) \leq r+1\}.$$

These normalized recovery equations determine their exact supports, while the converse generally fails. The operational model is scalar exact repair: each participating helper supplies one symbol of $\mathbb{F}_q$. Array-code access, subpacketization, and repair bandwidth are different optimization problems [24, 26, 14].

If $A \subseteq E \setminus \{x\}$ is the surviving helper set, put

$$f_{x,r}(A) = 1 \quad \Longleftrightarrow \quad A \text{ contains a member of } \mathcal{H}_x^{(\leq r)}(C).$$

Under independent helper survival with probabilities (s_y), the bounded repair reliability is

$$R_{x,r}((s_y)) = \mathbb{E}[f_{x,r}].$$

It depends only on the minimal recovery sets, but it is not determined by their minimum cardinality. Section 7 shows that even the complete relative-weight hierarchy does not determine the corresponding bounded reliability law.

For several target coordinates, a recovery equation is a dual subspace rather than one dual word. Let $P \subseteq E$, $J = E \setminus P$, and let $G = (G_P \mid G_J)$ be a full-row-rank generator matrix. If $T \leq \operatorname{im} G_P$ is a subspace of target linear combinations, a normalized recovery system for T consists of linear maps

$$\alpha : T \longrightarrow \mathbb{F}_q^P, \qquad \beta : T \longrightarrow \mathbb{F}_q^J$$

such that

$$G_P \alpha = \mathrm{id}_T, \qquad G_J \beta = -\mathrm{id}_T\,.$$

For each $t \in T$, the vector $(\alpha(t), \beta(t))$ is then a dual word. The map α records the chosen target normalization and β records the helper coefficients.

The helper union of the system is the support of $\beta(T)$:

$$\mathrm{supp}(\beta(T)) = \{j \in J : \text{some } t \in T \text{ has } \beta(t)_j \neq 0\}.$$

Its helper cost is $|\mathrm{supp}(\beta(T))|$. We call a target subspace internally recoverable when it lies in

$$W_P = \operatorname{im} G_P \cap \operatorname{im} G_J.$$

Only this recovered subspace dimension is counted below. Relations supported entirely on P have recovered dimension zero and do not consume helpers.

As a brief illustration of the coefficient layer, minimum-weight equations already recover the whole dual of an MDS code.

Theorem 2.1 (Minimum-weight MDS recovery equations span the dual)**.** *Let $C \leq \mathbb{F}_q^E$ be an $[n,k]_q$ MDS code with $1 \leq k < n$, and let $x \in E$. Then*

$$\operatorname{span} \mathcal{P}_x^{(\leq k)}(C) = C^{\perp},$$

and their supports form the complete k-uniform clutter on $E \setminus \{x\}$.

Proof. The dual has dimension $n-k$ and distance $k+1$. For each $(k+1)$-set T, restriction $C^{\perp} \to \mathbb{F}_q^{E \setminus T}$ has a nonzero kernel, whose words have support exactly T by the distance bound. Thus all stated supports occur. Fix a $(k-1)$-set $Q \subseteq E \setminus \{x\}$ and take the normalized word on $\{x,t\} \cup Q$ for each $t \in E \setminus (\{x\} \cup Q)$. Their private coordinates t make these $n-k$ words independent, hence a basis of $C^{\perp}$. □

The support clutter in this theorem depends only on the MDS parameters. The normalized equations retain the represented dual code. This is the first instance of the support–coefficient distinction used in the transfer and separation results below.

3 Puncturing, shortening, and relative recovery costs

Continue with $G = (G_P \mid G_J)$ and set

$$U_P = \operatorname{im} G_P, \qquad W_P = U_P \cap \operatorname{im} G_J,$$

$$K_P = \ker G_J, \qquad D_P = G_J^{-1}(U_P).$$

Proposition 3.1 (Associated nested code pair)**.** *Restriction of G_J gives an exact sequence*

$$0 \longrightarrow K_P \longrightarrow D_P \xrightarrow{G_J} W_P \longrightarrow 0. \tag{1}$$

Proof. By definition, $K_P = \ker G_J$ is contained in $D_P = G_J^{-1}(U_P)$. The kernel of the restricted map is therefore K_P. Its image is

$$G_J(D_P) = \operatorname{im} G_J \cap U_P = W_P,$$

so the restricted map is onto W_P. □

The first isomorphism theorem applied to the restricted map gives $D_P / K_P \cong W_P$.

Proposition 3.2 (Canonical puncturing–shortening description)**.** *With puncturing and shortening taken onto the helper set J,*

$$D_P = \operatorname{punct}_J(I^{\perp}), \qquad K_P = \operatorname{short}_J(I^{\perp}). \tag{2}$$

Taking duals inside $\mathbb{F}_q^J$ gives

$$D_P^{\perp} = \operatorname{short}_J(I), \qquad K_P^{\perp} = \operatorname{punct}_J(I). \tag{3}$$

Consequently the nested pair depends only on the code I and the split $E = P \sqcup J$; changing the row basis of a generator matrix leaves it unchanged.

Proof. A vector $a \in \mathbb{F}_q^J$ lies in $\operatorname{punct}_J(I^{\perp})$ exactly when some $p \in \mathbb{F}_q^P$ satisfies $G_P p + G_J a = 0$, equivalently $G_J a \in \operatorname{im} G_P$, which is $a \in D_P$. It lies in $\operatorname{short}_J(I^{\perp})$ exactly when $(0,a) \in I^{\perp}$, equivalently $G_J a = 0$, which is $a \in K_P$. The dual identities follow from the standard puncturing–shortening duality. □

Thus a monomial relabeling of helpers gives the corresponding monomially equivalent pair, while a row-basis change does nothing to the pair itself.

For a subspace $A \leq \mathbb{F}_q^J$, write $\operatorname{supp} A$ for the union of the supports of its vectors. For nested linear codes $K \subseteq D \leq \mathbb{F}_q^J$, the tth relative generalized Hamming weight is

$$M_t(D, K) = \min\{|\operatorname{supp} L| : L \leq D, \dim L - \dim(L \cap K) = t\}.$$

Equivalently, one may minimize over subspaces disjoint from K and of dimension t. Strict growth and the relative Singleton bound are Proposition 2 and Theorem 4 of Luo et al. [16]; standard treatments also cover ordinary Wei duality and computational forms [9, 23]. For nested code pairs in linear secret sharing, RDLPs and RGHWs already give exact information and reconstruction thresholds [13].

The following result is the standard nested-code support invariant applied to the canonical puncturing–shortening pair, with the normalization and helper support correspondence stated in recovery language.

Theorem 3.3 (Relative weights are exact recovery costs)**.** *Let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The minimum helper-union size among recovery systems for all internally recoverable t-dimensional subspaces of target linear combinations is*

$$\mu_t(P) = M_t(D_P, K_P).$$

More precisely, normalized recovery systems of recovered dimension t are in support-preserving correspondence, after forgetting the target normalization, with subspaces

$$L \leq D_P, \qquad \dim L = t, \qquad L \cap K_P = 0,$$

while the target normalization is a linear section through G_P.

Proof. Let a normalized system recover a t-dimensional subspace $T \leq W_P$. In the notation of Section 2, put $L = \beta(T)$. Since $G_J\beta = -\operatorname{id}_T$, the map β is injective, $L \leq D_P$, and $L \cap K_P = 0$. Moreover $G_J L = T$, and the union of helpers in the system is exactly $\operatorname{supp} L$.

Conversely, let $L \leq D_P$ have dimension t and meet K_P trivially. Then $G_J|_L$ is an isomorphism from L onto the t-dimensional subspace $T = G_J L \leq W_P$. Choose a linear section $\alpha : T \to \mathbb{F}_q^P$ with $G_P\alpha = \operatorname{id}_T$. Put $\beta = -(G_J|_L)^{-1}$. Then $G_P\alpha + G_J\beta = 0$, so (α, β) is a normalized recovery system with helper union $\operatorname{supp} L$.

It remains to compare this minimum with the standard relative-weight definition. If $L' \leq D_P$ satisfies

$$\dim L' - \dim(L' \cap K_P) = t,$$

choose a vector-space complement L to $L' \cap K_P$ inside L'. Then $\dim L = t$, $L \cap K_P = 0$, and $\operatorname{supp} L \subseteq \operatorname{supp} L'$. The reverse inclusion of feasible classes is immediate. The two minima are equal. $\square$

The corresponding relative dimension/length profile is

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H| = s}} \big(\dim(D_P \cap \mathbb{F}_q^H) - \dim(K_P \cap \mathbb{F}_q^H)\big),$$

where $\mathbb{F}_q^H$ denotes vectors supported on H.

Proposition 3.4 (Recovery interpretation of the relative profile)**.** *For* $0 \le s \le |J|$*,*

$$K_s(D_P, K_P) = \max_{\substack{H \subseteq J \\ |H|=s}} \dim\bigl(W_P \cap \operatorname{span}(G_H)\bigr).$$

Consequently,

$$M_t(D_P, K_P) = \min\{s : K_s(D_P, K_P) \ge t\}.$$

Thus K_s *is the maximum number of independent target linear combinations recoverable from a fixed budget of* s *helpers.*

Proof. Fix H. Restrict the exact map (1) to $D_P \cap \mathbb{F}_q^H$. Its kernel is $K_P \cap \mathbb{F}_q^H$, and its image is $W_P \cap \operatorname{span}(G_H)$. Rank–nullity gives the first equality after maximizing over H. The second is the standard inverse relation between the relative dimension/length profile and relative generalized weights; it also follows directly from Theorem 3.3. □

The shortening–puncturing duality behind the relative profiles of nested codes is standard in wiretap and ramp-secret-sharing theory [16, Lemma 1 and Theorem 2]; see also [9, eqs. (1)–(2)]. Its recovery interpretation gives the failure-side counterpart of Theorem 3.3.

Proposition 3.5 (Failures and residual ambiguity)**.** *Let* $\ell = \dim W_P$*. By* (3)*, the dual nested pair is* $\operatorname{short}_J(I) \subseteq \operatorname{punct}_J(I)$*. If the helpers in* $H \subseteq J$ *survive and those in* $F = J \setminus H$ *fail, then the unrecovered target dimension is*

$$\ell - \dim\bigl(W_P \cap \operatorname{span}(G_H)\bigr) = \dim(K_P^\perp \cap \mathbb{F}_q^F) - \dim(D_P^\perp \cap \mathbb{F}_q^F). \tag{4}$$

Consequently, for $1 \le t \le \ell$*,*

$$M_t(K_P^\perp, D_P^\perp) = \min\left\{|F| : \dim\bigl(W_P \cap \operatorname{span}(G_{J\setminus F})\bigr) \le \ell - t\right\}. \tag{5}$$

Thus the dual relative weight is the minimum number of helper failures that leave at least t *target dimensions ambiguous.*

Proof. For a code $C \le \mathbb{F}_q^J$, puncturing onto F gives

$$\dim C - \dim(C \cap \mathbb{F}_q^H) = \dim(C|_F) = |F| - \dim(C^\perp \cap \mathbb{F}_q^F).$$

Apply this identity to D_P and K_P and subtract. Since $\dim D_P - \dim K_P = \ell$, Proposition 3.4 gives (4). The right-hand side is the relative dimension of the dual pair $D_P^\perp \subseteq K_P^\perp$ on F. Its least support size at value at least t is $M_t(K_P^\perp, D_P^\perp)$: a subspace attaining the relative weight is supported on such an F, while any relative dimension at least t admits a t-dimensional subspace disjoint from $D_P^\perp \cap \mathbb{F}_q^F$. This proves (5). □

The strict inequalities

$$1 \le M_1(D_P, K_P) < \cdots < M_\ell(D_P, K_P)$$

therefore say that each additional independent target combination requires a strictly larger helper union. The next section shows that concatenation adds the same inner-dual cost $d(I^\perp)$ to every level of this hierarchy.

4 Exact confinement under concatenation

Let $L/\mathbb{F}_q$ be a finite extension and let $\iota : L \xrightarrow{\sim} I \le \mathbb{F}_q^E$ be an $\mathbb{F}_q$-linear isomorphism, used as the inner encoder. Assume $0 < \dim I < |E|$. For an L-linear outer code $O \le L^N$, write

$$O \circ I = \{(\iota(a_1), \ldots, \iota(a_N)) : (a_1, \ldots, a_N) \in O\}.$$

The proof of confinement uses the linear map

$$\Phi_I : \mathbb{F}_q^E \longrightarrow L^*, \qquad \Phi_I(y)(a) = \langle y, \iota(a) \rangle.$$

Here $L^* = \operatorname{Hom}_{\mathbb{F}_q}(L, \mathbb{F}_q)$. Its kernel is $I^\perp$. A block vector $y = (y_1, \ldots, y_N)$ belongs to $(O \circ I)^\perp$ precisely when

$$(\Phi_I(y_1), \ldots, \Phi_I(y_N)) \in \mathrm{FD}(O), \tag{6}$$

where

$$\mathrm{FD}(O) = \{(\beta_j) \in (L^*)^N : \textstyle\sum_j \beta_j(a_j) = 0 \text{ for every } (a_j) \in O\}.$$

Indeed, both conditions say that y is orthogonal to every encoded outer word. Write $\beta_j(a) = \mathrm{Tr}_{L/\mathbb{F}_q}(b_j a)$. The functional-dual condition becomes

$$\mathrm{Tr}_{L/\mathbb{F}_q}\left(\sum_j b_j a_j\right) = 0 \qquad \text{for every } (a_j) \in O.$$

Because O is L-linear, the same holds after multiplying (a_j) by every $\lambda \in L$. Nondegeneracy of the trace pairing then forces $\sum_j b_j a_j = 0$. Hence $(b_j) \in O^\perp$, and the converse is immediate. This identification preserves block support, so the support distance of $\mathrm{FD}(O)$ is $d(O^\perp)$.

Fix a target block and a nonzero target-message subspace $T \le U_P$. Let $\rho_T(I)$ be the minimum helper-union size of a normalized recovery system for T in the inner code, with value ∞ when no such system exists. Thus $\rho_T(I) < \infty$ exactly when $T \le W_P$. In a concatenation, a normalized system for T means a linear family of dual equations whose target-block coefficient map $\alpha : T \to \mathbb{F}_q^P$ satisfies $G_P\alpha = \mathrm{id}_T$ under the fixed inner-message identification. This agrees with the intrinsic target image of the concatenated generator whenever the outer projection onto the chosen block is surjective. In particular, $d(O^\perp) > 1$ guarantees surjectivity: an L-linear coordinate image is either 0 or L, and a zero image would put a weight-one word in $O^\perp$. A system is *confined* if every one of its equations is supported in the target block.

For an $\mathbb{F}_q$-linear map $B : T \to L^*$, define

$$\lambda_{I,T}(B) = \min_{\substack{Y: T \to \mathbb{F}_q^E \text{ linear} \\ \Phi_I Y = B}} |\operatorname{supp} Y(T)|. \tag{7}$$

For the target block, define

$$\mu_{I,P,T}(B) = \min_{\substack{\alpha: T \to \mathbb{F}_q^P,\ \beta: T \to \mathbb{F}_q^J \text{ linear} \\ G_P\alpha = \mathrm{id}_T \\ \Phi_I(\alpha,\beta) = B}} |\operatorname{supp} \beta(T)|. \tag{8}$$

An empty minimum is ∞. After choosing a basis of T, (7) is the minimum union support of representatives of prescribed cosets of $I^\perp$. This is the fixed-instance quantity underlying generalized coset weights and generalized covering radii [27, 7]. The target version retains the normalization required by the recovery problem. In particular,

$$\mu_{I,P,T}(0) = \rho_T(I). \tag{9}$$

Let

$$\mathcal{B}_T(O) = \mathrm{Hom}_{\mathbb{F}_q}\big(T, \mathrm{FD}(O)\big).$$

For $B \in \mathcal{B}_T(O)$, write $B = (B_h)_{h=1}^N$ with $B_h : T \to L^*$. Assume $N \geq 2$, fix a target block j whose coordinate projection $O \to L$ is nonzero, hence surjective, and put

$$\begin{aligned}\Gamma_{j,T}(O, I) = \min\Bigg\{&\mu_{I,P,T}(0) + d(I^\perp), \\ &\min_{0 \neq B \in \mathcal{B}_T(O)} \Bigg(\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h)\Bigg)\Bigg\}.\end{aligned} \tag{10}$$

Theorem 4.1 (Exact finite confinement from prescribed coset supports)**.** *Under the hypotheses above, $\Gamma_{j,T}(O, I)$ is the least helper-union size of a nonconfined normalized recovery system for T in $O \circ I$, with value ∞ if no such system exists. Thus every cost-$\leq r$ system for T is confined if and only if*

$$r < \Gamma_{j,T}(O, I). \tag{11}$$

For $1 \leq t \leq \ell = \dim W_P$, set

$$\Gamma_{j,t}(O, I) = \min_{\substack{T \leq W_P \\ \dim T = t}} \Gamma_{j,T}(O, I). \tag{12}$$

Then every cost-$\leq r$ system for every internally recoverable t-dimensional target subspace is confined if and only if $r < \Gamma_{j,t}(O, I)$. Below either threshold, restriction and zero-extension are inverse bijections on normalized systems and preserve exact helper supports.

Proof. Write a recovery system as an $\mathbb{F}_q$-linear map $Y : T \to (O \circ I)^\perp$ and let Y_h be its block maps. By (6),

$$B_h = \Phi_I Y_h$$

defines a linear map $B : T \to \mathrm{FD}(O)$. Conversely, compatible linear lifts of the block maps of any such B give a map into the concatenated dual. At the target block, compatibility with the chosen target space is exactly the constraint in (8).

Suppose first that $B \neq 0$. No nonzero element of $\mathrm{FD}(O)$ is supported only at j: testing such an element against outer words whose jth coordinate ranges over L would force its jth functional to vanish. Hence some block map B_h with $h \neq j$ is nonzero, and every lift is nonconfined. Distinct inner blocks have disjoint helper coordinates, so the minimum helper union for this fixed B is

$$\mu_{I,P,T}(B_j) + \sum_{h \neq j} \lambda_{I,T}(B_h).$$

The block lifts can be chosen independently, so this lower bound is attained.

If $B = 0$, every block map takes values in $I^\perp = \ker \Phi_I$. The target block costs $\rho_T(I) = \mu_{I,P,T}(0)$. Nonconfinement requires a nonzero map $T \to I^\perp$ in another block. Its image has union support at least $d(I^\perp)$, and equality is attained by $u \mapsto f(u)v$, where $0 \neq f \in T^*$ and v is a minimum-weight word of $I^\perp$.

The zero and nonzero functional sectors are exhaustive, proving (10) and (11). Minimizing over t-dimensional T proves the ranked statement. Below the first nonconfined cost, restriction to block j is inverse to zero-extension, with coefficients and supports unchanged. ∎

4.1 Repeated concatenation

The ordinary prescribed-coset costs retain the intermediate labels needed at the next concatenation level. Target-normalized composition also retains the target image in each inner block. The resulting formulas are instances of variable elimination over the min-sum semiring [2]. This is also the algebraic form of passing symbol-indexed inner costs to an outer decoder, as in soft and generalized-concatenated decoding [11, 3].

Let $\mathbb{F}_q \subseteq L \subseteq M$ be a tower of finite fields. Let B be an $\mathbb{F}_q$-linear represented code with message space L, and let A be an L-linear represented code with message space M. After the trace identifications, write their dual restriction maps as

$$\phi_B : \mathbb{F}_q^{E_B} \longrightarrow L, \qquad \phi_A : L^{E_A} \longrightarrow M.$$

For an $\mathbb{F}_q$-space T and $b : T \to L$, put

$$\Lambda_{B,T}(b) = \min_{\phi_B Y = b} |\operatorname{supp} Y(T)|.$$

Define $\Lambda_{A,T}$ and $\Lambda_{A\circ B,T}$ similarly. If $J \subseteq E_B$, let $\Lambda_{B,J,T}$ denote the same minimum with Y supported on J and ϕ_B restricted to those coordinates.

Proposition 4.2 (Composition of prescribed-coset support costs)**.** *For every $\mathbb{F}_q$-linear map $c : T \to M$,*

$$\Lambda_{A\circ B,T}(c) = \min_{X:T\to L^{E_A} \ \phi_A X = c} \sum_{e\in E_A} \Lambda_{B,T}(X_e). \tag{13}$$

When $T \neq 0$, set

$$\delta_{B,T} = \min_{0\neq b:T\to L} \Lambda_{B,T}(b), \qquad R_{B,T} = \max_{0\neq b:T\to L} \Lambda_{B,T}(b).$$

For $c \neq 0$, the exact formula has the sharp coarsening

$$\delta_{B,T}\Lambda_{A,T}(c) \leq \Lambda_{A\circ B,T}(c) \leq R_{B,T}\Lambda_{A,T}(c). \tag{14}$$

Neither constant can be improved without further hypotheses. Suppose that a target set in $A \circ B$ meets block e in $P_e \subseteq E_B$, and put $J_e = E_B \setminus P_e$ and $U_{P_e} = \operatorname{im} \phi_{B,P_e}$. If $T \leq M$ and $\iota_T : T \hookrightarrow M$ is the inclusion, then the target-normalized cost satisfies

$$\mu_{A\circ B,P,T}(c) = \min_{\substack{U,X:T\to L^{E_A} \ U_e(T)\subseteq U_{P_e} \ (e\in E_A) \\ \phi_A U = \iota_T,\ \phi_A X = c}} \sum_{e\in E_A} \Lambda_{B,J_e,T}(X_e - U_e). \tag{15}$$

Moreover,

$$d((A\circ B)^\perp) = \min\left\{ d(B^\perp), \min_{0\neq z\in A^\perp} \sum_{e\in E_A} \Lambda_{B,\mathbb{F}_q}(u \mapsto u z_e) \right\}. \tag{16}$$

These formulas iterate associatively. Substitution into (10) therefore gives the exact nonconfinement cost through any finite tower, with the zero and nonzero functional sectors kept separate at every level.

Proof. A leaf-level lift $Y : T \to \mathbb{F}_q^{E_A\times E_B}$ determines intermediate maps $X_e = \phi_B Y_e$. Its support is the disjoint union of its supports in the inner blocks. At fixed X, the block lifts are independent and attain their minima separately, which proves (13). Every nonzero intermediate coordinate map costs between $\delta_{B,T}$ and $R_{B,T}$. Minimizing its number of nonzero coordinates proves (14); a one-coordinate outer realization and a map attaining either extremum show that the constants are sharp.

For the target formula, let $a_e : T \to \mathbb{F}_q^{P_e}$ be the target coefficients and set $U_e = \phi_{B,P_e} a_e$. Every map into U_{P_e} has such a linear lift. Target normalization is exactly $\phi_A U = \iota_T$. If X_e is the total intermediate map in block e, its helper contribution must lift $X_e - U_e$ through ϕ_{B,J_e}, proving (15). Before taking minimum supports, retain the actual target maps a_e and the full helper lift sets. These compose by fiber product over the intermediate total maps X, so the same argument also transports every coefficient-labelled equation and its support.

A word in $(A \circ B)^\perp$ either has zero intermediate label, when a minimum word of $B^\perp$ in one block is optimal, or has a nonzero label $z \in A^\perp$, when the block representatives minimize independently. This proves (16). At three or more levels, either parenthesization minimizes over the same intermediate arrays and sums the same leaf-block costs, proving associativity and the final assertion. □

The trace identifications in this proposition are compatible with the tower. Indeed, trace transitivity gives

$$\sum_e \mathrm{Tr}_{L/\mathbb{F}_q}\big(X_e(v)\,\iota_A(m)_e\big) = \mathrm{Tr}_{M/\mathbb{F}_q}\big(\phi_A(X(v))m\big),$$

and nondegeneracy of the trace pairing identifies the composite restriction map with $\phi_A \circ (\phi_B)_e$.

The scalar persistent obstruction alone is not compositional. Over $\mathbb{F}_2 \subseteq \mathbb{F}_4$, write the nonzero elements as $1, \omega, \omega^2 = 1 + \omega$ and take inner generator columns

$$I_1 : (1, 1, \omega), \qquad I_2 : (1, 1, \omega^2),$$

with the first coordinate targeted. Both have target recovery cost 1, dual distance 2, and hence persistent value 3. Their ordinary coset costs on $(1, \omega, \omega^2)$ are $(1, 1, 2)$ and $(1, 2, 1)$; their normalized target costs are $(0, 2, 1)$ and $(0, 1, 2)$. For an outer functional dual spanned by $(1, \omega)$, the nonzero-functional minima are respectively 1 and 2. Thus the labelled prescribed-coset functions, rather than one persistent minimum, are the closed numerical data under repeated concatenation.

The complete bounded structure across all recoverable ranks has a single bottleneck. If a system on $0 \neq T \leq W_P$ is nonconfined, some external block map is nonzero on a vector $u \in T$; restriction to $\langle u \rangle$ is still nonconfined and cannot enlarge its helper union. Consequently

$$\min_{0 \neq T \leq W_P} \Gamma_{j,T}(O, I) = \Gamma_{j,1}(O, I). \tag{17}$$

Thus every cost-$\leq r$ system for every internally recoverable target subspace is confined if and only if $r < \Gamma_{j,1}(O, I)$. This is the exact finite local-to-global gate for the complete bounded recovery data of the target/helper split.

The exact optimization also recovers the persistent obstruction without choosing a radius. Indeed, every nonzero functional sector costs at least $d(O^\perp) - 1$, while the zero-functional sector attains $M_t(D_P, K_P) + d(I^\perp)$ after minimization over T. Consequently,

$$d(O^\perp) \geq M_t(D_P, K_P) + d(I^\perp) + 1 \quad \Longrightarrow \quad \Gamma_{j,t}(O, I) = M_t(D_P, K_P) + d(I^\perp). \tag{18}$$

Theorem 4.3 (Confinement for a fixed target subspace)**.** *Let r be a nonnegative integer and let $O \leq L^N$ be an L-linear outer code with $N \geq 2$ and $d(O^\perp) > r + 1$, and assume $T \leq W_P$. Then $O \circ I$ confines every normalized recovery system for T of cost at most r if and only if*

$$r < \rho_T(I) + d(I^\perp). \tag{19}$$

Below this bound, zero-extension gives a support-preserving bijection between the inner systems of cost at most r and the corresponding systems in the concatenation. Hence the same conclusions hold for all sufficiently long members of any outer family with dual distance tending to infinity, at every fixed r.

Proof. If $0 \neq B \in \mathcal{B}_T(O)$, choose $u \in T$ with $B(u) \neq 0$. The tuple $B(u) \in \mathrm{FD}(O)$ has at least $d(O^\perp)$ nonzero blocks. At most one is the target block, so every lift has at least $d(O^\perp) - 1 > r$ helpers outside it. Thus no nonzero-functional sector in (10) occurs at cost at most r.

The zero-functional sector has exact cost $\mu_{I,P,T}(0) + d(I^\perp) = \rho_T(I) + d(I^\perp)$. Theorem 4.1 now gives both directions of (19) and the support-preserving bijection. $\square$

Minimizing the fixed-subspace threshold over all recovered subspaces of fixed dimension gives the promised hierarchy.

Theorem 4.4 (Confinement by recovered dimension after the outer-distance gate)**.** *Let r be a nonnegative integer, let $O \leq L^N$ be L-linear with $N \geq 2$ and $d(O^\perp) > r + 1$, and let $\ell = \dim W_P$ and $1 \leq t \leq \ell$. The concatenation $O \circ I$ confines every cost-$\leq r$ recovery system for every internally recoverable t-dimensional target subspace if and only if*

$$r < M_t(D_P, K_P) + d(I^\perp). \tag{20}$$

When (20) holds, restriction and zero-extension preserve every normalized equation and its exact helper support. For an outer family with dual distance tending to infinity, these conclusions hold for all sufficiently long members at every fixed r.

Proof. By Theorem 3.3, every such target subspace has inner cost at least $M_t(D_P, K_P)$, and some target subspace attains this minimum. Apply Theorem 4.3 to each subspace. The lower bound gives sufficiency uniformly; an attaining subspace and the external rank-one perturbation in that proof give necessity. $\square$

4.2 One-coordinate specialization

For $\beta \in L^*$, put

$$\lambda_I(\beta) = \min_{\Phi_I(y)=\beta} \mathrm{wt}(y), \qquad \mu_{I,x}(\beta) = \min_{\substack{\Phi_I(y)=\beta \\ y_x \neq 0}} \mathrm{wt}(y),$$

with value ∞ when a constrained fiber is empty. The first quantity is the usual coset-leader weight for the inner encoder, while the second retains the target-coordinate constraint. The functional-dual decomposition itself is classical concatenation machinery [5, Theorem 2.1].

Theorem 4.5 (Exact weighted finite-length confinement at one coordinate)**.** *Fix an outer block j, an inner coordinate x, and $N \geq 2$, and assume that the coordinate projection $O \to L$ at block j is nonzero, hence surjective. The minimum total Hamming weight of a word in $(O \circ I)^\perp$ that is nonzero at (j, x) and is not the zero-extension of a word of $I^\perp$ in block j is*

$$Z_{j,x}(O, I) = \min \left\{ \mu_{I,x}(0) + d(I^\perp), \min_{\substack{\beta=(\beta_h)\in\mathrm{FD}(O) \\ \beta\neq 0}} \left(\mu_{I,x}(\beta_j) + \sum_{h\neq j} \lambda_I(\beta_h) \right) \right\}. \tag{21}$$

A minimum over an empty set is ∞. Consequently, for every $r \geq 0$, restriction to block j and zero-extension are inverse, support-preserving bijections between the normalized recovery equations through x in I and those through (j, x) in $O \circ I$, both using at most r helpers, if and only if

$$Z_{j,x}(O, I) > r + 1. \tag{22}$$

The same equivalence holds for the exact helper-support families. When these conditions hold, the inclusion-minimal recovery sets are copied as well.

Proof. Write a concatenated dual word in blocks and apply (6). For a fixed nonzero functional tuple, the block representatives minimize independently, giving the second term of (21). For the zero tuple, nonconfinement costs $\mu_{I,x}(0)$ in the target block and $d(I^{\perp})$ in another block. The projection hypothesis makes confined words exactly the zero-extensions of inner-dual words, so these two sectors prove (21).

If the target column is nonzero, take $P = \{x\}$ and let $T \leq U_P$ be its span. After choosing a nonzero vector of T, a linear map $T \to L^*$ is one functional. Normalizing its target coefficient and rescaling the functional tuple identify the minima in (10) with those in (21). The target coordinate contributes one to total word weight and is not a helper, so

$$Z_{j,x}(O, I) = \Gamma_{j,T}(O, I) + 1.$$

Theorem 4.1 gives the claimed gate. If the target column is zero, the same gate follows directly from the exact minimum already proved. In either case, below the minimum every equation is confined, so restriction and zero-extension are inverse. In one dimension every exact helper support is the support of its normalized equation; the equivalence also holds for exact helper-support families and their inclusion-minimal members. □

Theorem 4.5 retains information that the outer support distance discards. The following family makes the difference strict without using a geometric inner code.

Proposition 4.6 (A family from a Singer cycle beyond the outer-distance gate)**.** *Let $k \geq 2$ and suppose*

$$\frac{q^k - 1}{q - 1} > (k+1)^2. \tag{23}$$

Let I be the $[k+1, k, 2]_q$ single-parity-check code generated by

$$G = (I_k \mid \mathbf{1}),$$

and identify its message space with $L = \mathbb{F}_{q^k}$. For every $k + 1 \leq n \leq 2k + 1$, there is an L-linear length-n outer code O with $d(O^{\perp}) = n$ such that (22) holds at helper radius $n - 1$ for every inner coordinate and every outer block. Thus all normalized radius-$(n-1)$ recovery equations and their exact helper supports are copied blockwise although the ordinary outer-distance gate fails.

In particular, this holds for the $[4, 3, 2]_5$ inner code and a length-five outer code at helper radius four.

Proof. The $k+1$ coordinate functionals of the encoder give a set S of $k+1$ points in the projective space on L^*. The multiplication maps of L, modulo $\mathbb{F}_q^*$, induce on that projective space a Singer cycle of order $(q^k - 1)/(q-1)$, acting regularly on its points [25]. At most $|S|^2 = (k+1)^2$ elements of the cycle send a point of S to a point of S, so (23) permits a multiplication map $a : L \to L$, explicitly $a(u) = \gamma u$ for some $\gamma \in L^{\times}$, whose dual action satisfies $a^*(S) \cap S = \varnothing$.

Set

$$O = \{u \in L^n : u_0 + a(u_1) + u_2 + \cdots + u_{n-1} = 0\}.$$

Every functional-dual tuple has the form $(f, f \circ a, f, \ldots, f)$. If $f \neq 0$, all n entries are nonzero, so $d(O^{\perp}) = n$. A realization of total weight n would have weight one in every block. Weight-one realizations are exactly the scalar multiples of the $k+1$ coordinate functionals, so both $[f]$ and $[f \circ a]$ would lie in S, contrary to the choice of a. Every nonzero tuple therefore costs at least $n+1$.

Every outer coordinate projection is surjective. Finally, $I^\perp$ is spanned by $(1,\dots,1,-1)$, so $d(I^\perp) = \mu_{I,x}(0) = k+1$ for every x. The zero-functional cost in (21) is $2k+2 > n$, and the nonzero-functional cost is at least $n+1$. Thus $Z_{j,x}(O,I) > n$ for every j,x. At $r = n-1$ the ordinary gate would require $d(O^\perp) > n$, which is false, while the exact weighted gate holds. Taking $(q,k,n) = (5,3,5)$ gives the stated concrete instance. Since the inner recovery equations have k helpers and $n-1 \geq k$, the asserted transfer is nonvacuous throughout the family. ☐

For a nondegenerate singleton target $P = \{x\}$, the first relative weight is the least number of helpers in a dual word through x. If the total-weight threshold used for one-coordinate equations is denoted by $z_x(I)$, then

$$z_x(I) = M_1(D_P, K_P) + 1 + d(I^\perp). \tag{24}$$

The helper-radius condition (20) is therefore $r+1 < z_x(I)$, with no change of convention hidden in the reduction. Theorem 4.1 is the common finite formula. The outer-distance gate removes its nonzero-functional sector and gives the RGHW threshold in every recovered dimension. Restricting instead to one target coordinate gives Theorem 4.5 without the gate.

5 Exact recovery optimization

Proposition 4.2 is constructive. It does not merely identify a confinement threshold: it gives a finite min–sum compiler whose states are the intermediate functionals that scalar summaries discard. The companion software, *ERGO-Comp* (Exact Recovery and Generalized-weight Optimization Compiler), implements this construction and retains coefficient-level witnesses for every optimum.

We first state the algorithm in flattened $\mathbb{F}_q$-linear coordinates. Let T have dimension t, let the outer message space have dimension m, and let the inner label space have dimension s. For the ith outer coordinate, write

$$A_i : \operatorname{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^s) \longrightarrow \operatorname{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$$

for its contribution to the outer restriction map, and let $\lambda_i(b)$ be the prescribed-coset support cost of the inner label b. Different inner blocks may have different cost functions.

Proposition 5.1 (Exact hierarchical recovery optimizer)**.** *For $c \in \operatorname{Hom}_{\mathbb{F}_q}(T, \mathbb{F}_q^m)$, define*

$$D_0(c) = \begin{cases} 0, & c = 0, \\ +\infty, & c \neq 0, \end{cases} \qquad D_i(c) = \min_b \bigl(D_{i-1}(c - A_i b) + \lambda_i(b)\bigr). \tag{25}$$

Then $D_N(c)$ is the exact prescribed-coset support cost of the concatenated code at label c. Storing a minimizing label b and predecessor state at each finite entry returns an exact block-labelled recovery witness, not only its scalar cost. The same recursion applies to the target-normalized formula after adjoining its target-image labels to the state.

If every label is enumerated, put $Q = q^{mt}$ and $S = q^{st}$. The direct table algorithm uses $O(NQS)$ min–sum transitions, $O(Q)$ working cost memory, and $O(NQ)$ predecessor entries when all witnesses are retained. Unreachable labels, quotient coordinates, projectively equivalent columns, and repeated block types may be compiled out without changing the optimum or the replayed witness.

Proof. After i blocks, $D_i(c)$ is the least sum of inner support costs among labels $(b_1,\dots,b_i)$ satisfying $\sum_{j\leq i} A_j b_j = c$. This invariant is true at $i=0$. Conditioning on the last label b_i gives exactly (25), so induction proves the claim. The leaf supports lie in disjoint blocks, hence their costs add. Backtracking the stored minimizers reconstructs the labels; the inner prescribed-coset witnesses then reconstruct the coefficient-level recovery system. There are Q states and S candidate labels per block, which gives the stated bounds. The target-normalized case is the same invariant applied to the paired labels in (15). □

The resulting table supplies all exact helper costs needed by the finite confinement formula, including its zero and nonzero outer-functional sectors. It can therefore emit a replayable recovery witness when a threshold fails, or a complete exhausted-state certificate when the requested label is unreachable. Iterating the compiler through a tower preserves the intermediate labels required at the next level; collapsing each table to one minimum would destroy precisely this compositional information.

The same support functions also lead to an exact capacity-aware repair scheduler. Suppose demand i has at most A admissible recovery choices. Choice a consumes a nonnegative integer load vector $\ell_{i,a} \in \mathbb{Z}_{\geq 0}^{H}$ on H helpers, and capacities are $\boldsymbol{c} = (c_1, \ldots, c_H)$. A demand may be skipped. This model distinguishes support occupancy from coefficient-weighted download and permits different capacities at different helpers.

Proposition 5.2 (Exact capacitated repair scheduling)**.** *Let $\mathcal{S} = \prod_{h=1}^{H}\{0, \ldots, c_h\}$. A dynamic program indexed by the current load $u \in \mathcal{S}$, with transitions $u \mapsto u + \ell_{i,a}$ whenever the result is within capacity and with a skip transition, returns the maximum number of simultaneously served demands and an exact assignment witness. It uses*

$$O\left(DAH\prod_{h=1}^{H}(c_h+1)\right) \quad \textit{time and} \quad O\left(\prod_{h=1}^{H}(c_h+1)\right) \quad \textit{working memory,} \tag{26}$$

apart from stored predecessors.

If there are positive integer weights w_h and a common $M > 0$ such that $w \cdot \ell_{i,a} = M$ for every admissible choice, a state serving exactly r demands lies on the graded shell

$$\mathcal{S}_r = \{u \in \mathcal{S} : w \cdot u = rM\}.$$

Thus the full box may be replaced by the union of its reachable shells. Two distinct states on one shell are incomparable in the componentwise order, so ordinary Pareto-dominance pruning cannot remove either one. The grading is therefore both an exact compression and a certificate of the limit of that common pruning rule.

Proof. After processing the first i demands, associate to every reachable load the largest served count and a predecessor realizing it. The skip and assignment transitions enumerate exactly the feasible decisions for demand $i+1$; induction gives optimality and witness recovery. Scanning at most A choices across H coordinates for every demand and box state gives (26). Under the grading hypothesis, additivity places every r-repair load on $\mathcal{S}_r$. If $u \leq v$ componentwise and both lie on the same shell, positivity of w forces $u = v$, proving the antichain assertion. □

5.1 Measured solver boundary

The specialized dynamic programs are most useful when compilation exposes a small algebraic state space. They are not replacements for a general constraint solver when the residual model contains unrelated side constraints. The fair deployment boundary is therefore:

$$\text{algebraic compilation} \longrightarrow \begin{cases} \text{specialized exact DP,} & \text{when the compiled state is sufficient,} \\ \text{CP-SAT,} & \text{for the remaining general constraints.} \end{cases}$$

Direct CP-SAT is an appropriate end-to-end baseline. A second, *structured CP-SAT* baseline receives the same symmetry reduction and algebraic preprocessing as ERGO-Comp, isolating the value of the specialized state engine from the value of compilation itself.

On the recorded x86-64 benchmark host, eleven interleaved rounds gave the following medians. These are implementation measurements, not asymptotic claims; exact inputs, outputs, tool versions, and replay metadata are in `algorithms/evidence/benchmarks.json`.

instance	ERGO-Comp	direct CP-SAT	structured CP-SAT	direct/ ERGO	structured/ ERGO
large box	71.88 μs	178.63 μs	177.22 μs	2.49	2.47
balanced shell	3.71 μs	1.143 ms	0.996 ms	308.04	268.44
small state	7.30 μs	2.753 ms	2.723 ms	376.98	372.85
large nonuniform	52.85 μs	1.250 ms	1.041 ms	23.65	19.70

Table 1: Exact capacitated-repair scheduling. Both CP-SAT columns return the same optimum and witness as ERGO-Comp; the structured column shares its safe preprocessing.

The largest gains occur on graded shells and other low-dimensional reachable state sets hidden inside a large capacity box. The practical companion is therefore useful both as a complete solver for structured recovery instances and as a certified front end that shrinks a residual CP-SAT model while retaining exact recovery witnesses and confinement thresholds.

6 Consequences of the threshold

The fixed-target relative weights sit between ordinary generalized weights and worst-case cooperative locality. Let $d_e(C^\perp)$ denote the eth generalized Hamming weight of $C^\perp$. For an e-set $P \subseteq E$, let $\kappa_C(P)$ be the least $|H|$ for which some e-dimensional subcode $A \le C^\perp$, supported on $P \cup H$, restricts isomorphically to $\mathbb{F}_q^P$. Set $\kappa_C(P) = \infty$ when no such subcode exists. Thus $\kappa_C(P)$ is the minimum number of helpers needed to recover all coordinates in P simultaneously.

Theorem 6.1 (Best-target generalized weight and first escape cost)**.** *For $1 \le e \le \dim C^\perp$,*

$$\min_{\substack{P \subseteq E \\ |P|=e}} \kappa_C(P) = d_e(C^\perp) - e. \tag{27}$$

Consequently, if every e-set is recoverable and $r_e^{\mathrm{coop}}(C) = \max_{|P|=e} \kappa_C(P)$, then

$$r_e^{\mathrm{coop}}(C) \ge d_e(C^\perp) - e.$$

For a fixed inner code in the eventual concatenation regime of Theorem 4.3, the least first nonconfinement cost over all e-coordinate target sets is

$$d_e(C^\perp) - e + d(C^\perp). \tag{28}$$

Proof. A normalized identity system on P spans an e-dimensional subcode of $C^\perp$ whose support is P together with its helper union. Hence $d_e(C^\perp) \le e + \kappa_C(P)$.

Conversely, let $A \le C^\perp$ have dimension e and support size $d_e(C^\perp)$. An information set P of a generator matrix of A has size e, and restriction $A \to \mathbb{F}_q^P$ is an isomorphism. Row reduction on these columns gives normalized identity equations whose helper union is $\operatorname{supp}(A) \setminus P$. This proves (27); the cooperative-locality inequality follows by taking the maximum instead of the minimum.

It remains to justify the escape formula when the target columns indexed by P are dependent. Put $U = \operatorname{im} G_P$. A normalized identity system on $\mathbb{F}_q^P$ and a normalized recovery system on U have the same minimum helper union. Indeed, restricting an identity system along a linear section $\alpha : U \to \mathbb{F}_q^P$ gives a system on U. Conversely, given a system on U, write

$$a = \alpha(G_P a) + \bigl(a - \alpha(G_P a)\bigr) \qquad (a \in \mathbb{F}_q^P).$$

The second summand lies in $\ker G_P$ and hence is a target-only dual equation, so adjoining these kernel equations recovers the full identity without adding helpers. Thus the common minimum is $\kappa_C(P)$.

Now repeat the block-functional argument of Theorem 4.3. Once the outer-functional case is excluded, the target block of any identity system uses at least $\kappa_C(P)$ helpers. If the system is nonconfined, a nonzero equation map in another block has union support at least $d(C^\perp)$; the two supports are disjoint. Conversely, attach a minimum-weight word of $C^\perp$ in a second block through any nonzero linear functional on $\mathbb{F}_q^P$. Hence the first nonconfinement cost for this P is $\kappa_C(P) + d(C^\perp)$. Minimizing over P and using (27) proves (28). □

Put $k = \dim I$ and $b = \operatorname{rank} G_J$. The exact sequence and the relative and ordinary Singleton bounds give

$$M_t(D_P, K_P) \leq b - \ell + t, \qquad d(I^\perp) \leq k + 1. \tag{29}$$

Theorem 6.2 (MDS thresholds and rigidity)**.** *For $1 \leq t \leq \ell$,*

$$M_t(D_P, K_P) + d(I^\perp) \leq 2k - \ell + t + 1. \tag{30}$$

Suppose I is MDS, and put

$$u = \min\{k, |P|\}, \qquad b = \min\{k, |J|\}.$$

Then $\ell = u + b - k$. If $\ell \geq 1$, then, for $1 \leq t \leq \ell$,

$$M_t(D_P, K_P) = k - u + t, \qquad M_t(D_P, K_P) + d(I^\perp) = 2k - u + t + 1. \tag{31}$$

If $|J| \geq k$, then $\ell = u$ and these values attain the ceiling in (30) at every rank. Conversely, when $\ell \geq 1$, equality in (30) at $t = 1$ forces I to be MDS and forces equality at every rank.

Proof. The ceiling follows from (29) and $b \leq k$. For an MDS code the column matroid is uniform, so $\operatorname{rank} G_P = u$, $\operatorname{rank} G_J = b$, $\ell = u + b - k$, and, for $H \subseteq J$ with $|H| = h \leq k$,

$$\dim\bigl(\operatorname{span} G_P \cap \operatorname{span} G_H\bigr) = \max\{0, u + h - k\}.$$

Thus Proposition 3.4 gives $M_t = k - u + t$ for $1 \leq t \leq \ell$; here $k - u + t \leq b \leq |J|$. Now $d(I^\perp) = k + 1$, and if $|J| \geq k$, then $b = k$ and $\ell = u$, proving the remaining direct assertions.

Equality at $t = 1$ forces equality in both Singleton bounds and $b \leq k$. Hence $d(I^\perp) = k + 1$, so I is MDS, and $M_1 = b - \ell + 1$. Strict growth then gives $M_t \geq b - \ell + t$, while relative Singleton gives the reverse inequality; every rank is extremal. □

We next record the global parameters and the density of the transferred data. Let the inner code have parameters $[m, k, d]_q$, so $[L : \mathbb{F}_q] = k$. Let $K_N = \dim_L O_N$ and $D_N = d(O_N)$.

Corollary 6.3 (Positive-density realization)**.** *Assume $d(O_N^\perp) \to \infty$. Below any fixed threshold supplied by Theorem 4.3 or Theorem 4.4, for all sufficiently large N every inner normalized recovery system and exact helper support is copied in each of the N blocks. Each fixed inner coordinate type has density $1/m$ among the concatenated coordinates; the union of the types in P has density p/m.*

In particular, if $r < M_1(D_P, K_P) + d(I^\perp)$, then all cost-$\leq r$ systems for all nonzero subspaces of W_P are copied simultaneously in every block. Hence the complete bounded local recovery data through radius r occur on the stated positive-density coordinate classes.

The concatenated code has length mN, dimension kK_N, and minimum distance at least dD_N. If

$$\frac{K_N}{N} \longrightarrow R > 0, \qquad \liminf_N \frac{D_N}{N} \geq \delta > 0,$$

then the concatenated family has rate kR/m and relative distance at least $d\delta/m$. Outer families satisfying all three hypotheses exist over every finite field L.

Proof. The transfer statement is the zero-extension bijection in the confinement theorems, applied independently in every block. There are mN coordinates, and each inner coordinate occurs once in each block, which gives the density claims. The blockwise inner encoder is injective, so restriction of scalars gives dimension kK_N. A nonzero outer word has at least D_N nonzero symbols, each encoding to an inner word of weight at least d. This proves the distance and asymptotic assertions.

For the existence statement, write $Q = |L|$. Choose $0 < R < 1$ and positive $\delta, \delta^\perp$ such that

$$H_Q(\delta) < 1 - R, \qquad H_Q(\delta^\perp) < R,$$

where H_Q is the Q-ary entropy function. The first-moment argument for a random K_N-dimensional L-linear code bounds the expected numbers of nonzero primal and dual words below the two cutoffs, up to subexponential factors, by

$$Q^{N(H_Q(\delta)+R-1)} \qquad \text{and} \qquad Q^{N(H_Q(\delta^\perp)-R)},$$

respectively. Both tend to zero by the displayed inequalities, so one may choose a sequence with rate tending to R, primal relative distance at least δ, and dual relative distance at least $\delta^\perp$; see also [17, Chapter 17]. □

Exact support transfer also preserves the fractional load-balancing region used in service-rate problems. We state the consequence because it requires only the inclusion-minimal recovery sets, whereas the theorem transfers all normalized coefficients as well.

Fix one inner target block with helper set J. Let $\mathcal{A}$ be a finite set of demands on that block, with demand a represented by a nonzero target subspace $T_a \leq W_P$, and let $\mathcal{M}_a^{(\leq r)}$ be the inclusion-minimal exact helper supports of size at most r for $a \in \mathcal{A}$. For a nonnegative helper-capacity vector c, define

$$\Lambda_{\mathcal{A}}^{(\leq r)}(c) = \left\{ \lambda \geq 0 : \begin{array}{l} \text{there are } f_{a,R} \geq 0 \text{ with } \lambda_a = \sum_{R \in \mathcal{M}_a^{(\leq r)}} f_{a,R}, \\ \displaystyle\sum_a \sum_{R \ni j} f_{a,R} \leq c_j \text{ for every helper } j \end{array} \right\}.$$

This is the bounded service-rate region of the recovery-set system [19, 6].

Corollary 6.4 (Transfer of bounded service-rate regions)**.** *Let $O \leq L^N$ be an outer code with $N \geq 2$ whose projection onto the target block is nonzero. Suppose*

$$r < \Gamma_{j,T_a}(O, I) \qquad (a \in \mathcal{A}).$$

Transport capacities from the inner helper set to its copy in that block of $O \circ I$. Then the inner and concatenated bounded service-rate regions are equal. Capacities assigned outside the target block do not change this equality. In particular, the hypothesis follows from $d(O^{\perp}) > r + 1$ and the corresponding inner inequalities. Consequently, for an outer family with $d(O_N^{\perp}) \to \infty$, the equality holds for all sufficiently large N whenever those inner inequalities hold.

Proof. Theorem 4.1 confines every cost-$\leq r$ system for every demand. Thus restriction and zero-extension copy every inclusion-minimal support and create no cross-block minimal support. The helper bijection therefore transports the feasible variables $f_{a,R}$ and every capacity inequality in both directions. Allowing the full upward-closed recovery-set family gives the same region: flow assigned to a nonminimal set can be moved to a contained minimal set and weakly decreases every helper load. Thus cross-block supersets and capacities outside the target block are irrelevant. $\square$

7 What the relative-weight hierarchy forgets

Relative generalized weights record minimum support sizes. They do not record the overlap pattern of all recovery sets, which controls stochastic repair. Nor does the abstract pair record its ambient inner-code realization, which can change $d(I^{\perp})$ and the prescribed-coset support costs in (10). The information loss used in this paper is

$$\text{normalized recovery systems} \longrightarrow \text{exact helper supports} \longrightarrow \text{relative-weight minima.}$$

The exact finite concatenation formula also needs the outer functional dual and the joint support costs of the prescribed inner cosets. The RGHW layer is the relative-matroid support minimum; the extra finite transfer data retain a chosen linear realization. We give one separation at each of the two losses relevant to reliability and confinement.

First consider a one-dimensional recovered space. On the helper set $\{0, \ldots, 8\}$, let

$$\mathcal{A} = \{158, 026, 045, 013, 478\},$$
$$\mathcal{B} = \{168, 237, 078, 015, 124\},$$

where, for example, 158 denotes $\{1, 5, 8\}$. Both are realizable as the minimum recovery sets at a distinguished coordinate of a represented $[10, 4, 6]_q$ code over a common finite prime field.

Proposition 7.1 (Rank-one reliability separation)**.** *The two distinguished coordinates above have the same minimum helper cost $M_1 = 3$. Under independent homogeneous helper survival with probability s, their radius-three repair reliabilities are*

$$R_{\mathcal{A}}(s) = 5s^3 - 7s^5 - s^6 + 5s^7 - s^9, \tag{32}$$
$$R_{\mathcal{B}}(s) = 5s^3 - 7s^5 - 2s^6 + 8s^7 - 3s^8. \tag{33}$$

In particular, the complete rank-one relative-weight hierarchy does not determine bounded repair reliability.

Proof. The five triples in each clutter are the minimum recovery sets, so both minimum costs are three. For $k = 1, \ldots, 5$, the numbers of k-edge subfamilies according to union size are

k	$\mathcal{A}$	$\mathcal{B}$
1	$5u^3$	$5u^3$
2	$7u^5 + 3u^6$	$7u^5 + 3u^6$
3	$2u^6 + 6u^7 + 2u^8$	$u^6 + 8u^7 + u^8$
4	$u^7 + 2u^8 + 2u^9$	$4u^8 + u^9$
5	u^9	u^9.

Inclusion–exclusion for the five edge-containment events gives (32) and (33).

We now give the representation argument. Over an infinite field K, choose five projective lines $\ell_1, \ldots, \ell_5$. For $\mathcal{A}$, make ℓ_2, ℓ_3, ℓ_4 concurrent and label

$$0 = \ell_2 \cap \ell_3 \cap \ell_4, \quad 1 = \ell_1 \cap \ell_4, \quad 4 = \ell_3 \cap \ell_5, \quad 5 = \ell_1 \cap \ell_3, \quad 8 = \ell_1 \cap \ell_5.$$

Choose $2, 6$ on ℓ_2, 3 on ℓ_4, and 7 on ℓ_5. For $\mathcal{B}$, make ℓ_1, ℓ_4, ℓ_5 concurrent at 1, put

$$8 = \ell_1 \cap \ell_3, \quad 7 = \ell_2 \cap \ell_3, \quad 0 = \ell_3 \cap \ell_4, \quad 2 = \ell_2 \cap \ell_5,$$

and choose $6, 3, 5, 4$ on $\ell_1, \ell_2, \ell_4, \ell_5$, respectively. At each free choice, avoid the finitely many unwanted joining lines. The displayed triples are then exactly the collinear triples.

Choose representatives $p_i \in K^3$ of either plane configuration, let $x = (1, 0, 0, 0)$, and lift helper i to $h_i = (t_i, p_i) \in K^4$. Dependence of the lifts over any displayed line and dependence of any four helper lifts are proper linear conditions on the t_i. For a displayed line, its three quotient vectors have a unique dependence up to scalar, and the same coefficients give one nonzero linear equation in the t_i. For four helper points, the quotient vectors span K^3, so their lifted determinant is a nonzero linear form in the four first coordinates. Avoiding this finite union makes every helper triple and quadruple independent, while the dependent four-sets through x are exactly $\{x\} \cup A$ for the five prescribed triples A. Choose the construction over $\mathbb{Q}$ and reduce both configurations modulo one prime avoiding the finitely many nonzero determinants. The two row-span codes then have rank four, dual distance four, and minimum recovery sets exactly $\mathcal{A}$ and $\mathcal{B}$. A hyperplane not containing x contains at most three helpers because every four helpers are independent. A hyperplane containing x contains at most one displayed collinear triple, and such a triple gives four columns in a hyperplane. Thus the maximum hyperplane intersection has size four, so both codes have distance $10 - 4 = 6$. □

The rank-one separation extends to every quotient rank without changing any relative generalized weight. For a nested pair $K \subseteq D \leq \mathbb{F}_q^J$ with $\dim(D/K) = \ell$, define its radius-R full-quotient reliability by

$$\Pr\{\exists L \leq D : \ L + K = D, \ |\operatorname{supp} L| \leq R, \ \operatorname{supp} L \subseteq S\},$$

where $S \subseteq J$ is the random set of independently surviving helpers, each present with probability s.

Theorem 7.2 (Equal relative weights, different reliability for full recovery)**.** *For every $\ell \geq 1$, there are two nested code pairs of quotient dimension ℓ with identical relative generalized Hamming weights $M_1, \ldots, M_\ell$ but different bounded reliability laws for recovery of the full quotient.*

Proof. For $\ell = 1$, use Proposition 7.1. For $\ell > 1$, let $K_0 \subset D_0$ be either of its two associated rank-one pairs. On disjoint helper blocks, add the same $\ell - 1$ pairs

$$0 \subset \langle \mathbf{1}_{L_i} \rangle \qquad (1 \leq i \leq \ell - 1),$$

where each padding line has the unique support of size L_i. Thus

$$K = K_0 \oplus 0 \oplus \cdots \oplus 0, \qquad D = D_0 \oplus \langle \mathbf{1}_{L_1} \rangle \oplus \cdots \oplus \langle \mathbf{1}_{L_{\ell-1}} \rangle.$$

For a direct sum of rank-one quotient pairs, the tth relative weight is the sum of the t smallest component costs. Both constructions therefore have the same hierarchy, obtained from the common multiset $\{3, L_1, \ldots, L_{\ell-1}\}$.

To verify the direct-sum formula, project a feasible helper subspace to each quotient component. A t-dimensional quotient image has nonzero projection to at least t of the one-dimensional components. On disjoint helper blocks, each active projection pays at least that component's minimum cost, so the total support is at least the sum of the t smallest costs. The direct sum of minimum words from the t cheapest components attains the bound.

Set $R = 3 + \sum_i L_i$. A recovery subspace of full quotient rank projects nontrivially to every padding component and hence uses its entire forced support. Removing those supports leaves a radius-three recovery set in the base component. Conversely, any base radius-three recovery set together with the padding generators recovers the full quotient at radius R. Thus

$$R_{\mathrm{full},R}(s) = s^{\sum_i L_i} R_{\mathrm{base},3}(s).$$

The two polynomials remain different by (32)–(33).

Every pair in the construction is realized by an inner code. Put $V = \mathbb{F}_q^J/K$, let $\pi : \mathbb{F}_q^J \to V$ be the quotient map, and choose an isomorphism $\alpha : \mathbb{F}_q^\ell \to D/K \leq V$. The generator map

$$G : \mathbb{F}_q^\ell \oplus \mathbb{F}_q^J \longrightarrow V, \qquad G(a, x) = \alpha(a) + \pi(x),$$

has helper kernel K and helper preimage of the target span equal to D. Its associated nested pair is therefore $K \subseteq D$. □

The second separation fixes the abstract nested pair itself. This is not a row-basis change of one inner code: by Proposition 3.2, such a change leaves (D_P, K_P) unchanged. Here a *coefficient presentation* means an ambient inner dual code, target coordinates, and target coefficient map that realize a prescribed abstract pair $K \subseteq D$ as its shortening–puncturing pair. Different such realizations may have different ambient dual distances. Let

$$K = 0, \qquad D = \langle u, v \rangle \leq \mathbb{F}_2^3, \qquad u = 100, \quad v = 011.$$

The nonzero helper words have weights $1, 2, 3$, so $(M_1, M_2) = (1, 3)$.

Proposition 7.3 (Different ambient realizations can change confinement)**.** *Present the target space first by*

$$10 \mapsto u, \qquad 01 \mapsto v, \qquad 11 \mapsto u + v,$$

and then by

$$11 \mapsto u, \qquad 10 \mapsto v, \qquad 01 \mapsto u + v.$$

The two inner codes have the same associated nested pair, the same unlabeled set of helper supports, and the same complete relative-weight hierarchy. Their inner-dual weight multisets are respectively $\{2, 3, 5\}$ *and* $\{3, 3, 4\}$*. Hence their values of* $M_t(D, K) + d(I^\perp)$*, equivalently their eventual confinement thresholds, are*

$$(\Gamma_1, \Gamma_2) = (3, 5) \qquad \text{and} \qquad (\Gamma_1, \Gamma_2) = (4, 6).$$

Proof. Use the graph-code realization

$$I^\perp = \{(-\alpha^{-1}(x), x) : x \in D\},$$

where α is the displayed identification of the two-dimensional target space with D. Adding the target weight to the helper weight gives the two listed multisets directly. Thus $d(I^\perp)$ is two in the first presentation and three in the second. Adding this distance to $(M_1, M_2) = (1, 3)$ gives the two threshold sequences by Theorem 4.4. □

8 The projective-simplex family

The relative-weight hierarchy can be computed exactly for a non-MDS family. Assume $m \geq 2$, put

$$N = \frac{q^m - 1}{q - 1},$$

and let A be an $m \times N$ matrix containing one representative of every point of $\mathrm{PG}(m-1, q)$. Define the inner code by

$$I_m^{\perp} = \{(a, aA) : a \in \mathbb{F}_q^m\}. \tag{34}$$

Take the first m coordinates as targets and the projective coordinates as helpers. The associated nested pair is isomorphic to

$$0 \subseteq S_m = \{aA : a \in \mathbb{F}_q^m\},$$

where S_m is the q-ary simplex code. The generalized-weight formula for S_m is classical; in fact every t-dimensional simplex subcode has the same support size [21]. We include the short projective count to connect that hierarchy to the confinement and reliability statements below.

Theorem 8.1 (Projective-simplex thresholds)**.** *For $1 \leq t \leq m$,*

$$M_t(S_m, 0) = \frac{q^m - q^{m-t}}{q - 1}, \tag{35}$$

and

$$d(I_m^{\perp}) = q^{m-1} + 1. \tag{36}$$

Hence

$$\Gamma_t := M_t(S_m, 0) + d(I_m^{\perp}) = \frac{q^m - q^{m-t}}{q - 1} + q^{m-1} + 1. \tag{37}$$

In outer families with dual distance tending to infinity, Γ_t is the eventual first nonconfined helper cost at recovered dimension t. The family is non-MDS for $m \geq 2$; for $m \geq 3$ the successive increments of Γ_t are unequal.

Proof. The target restriction of a word $(a, aA) \in I_m^{\perp}$ is a, and no nonzero dual word has zero target restriction. Hence normalized recovery systems correspond, with support preserved, to subcodes of S_m; this proves the asserted identification of the associated pair.

Let $U \leq \mathbb{F}_q^m$ have dimension t. The common zero coordinates of the subcode UA are the projective points in $U^{\perp}$, a subspace of projective dimension $m - t - 1$. Thus

$$|\operatorname{supp}(UA)| = N - \frac{q^{m-t} - 1}{q - 1} = \frac{q^m - q^{m-t}}{q - 1},$$

independently of U. This proves (35). Every nonzero simplex word aA has weight q^{m-1}, while a nonzero target vector a has weight at least one. Equality is attained by a coordinate vector, which proves (36). The confinement formula is Theorem 4.4. The non-MDS and increment assertions follow from the displayed parameters. $\square$

This family also gives an exact stochastic law at every recovered dimension. Let each helper survive independently with probability s, and let F be the set of failed projective points. The available recovery equations are indexed by

$$\{a \in \mathbb{F}_q^m : aA|_F = 0\} = \left(\operatorname{span} F\right)^{\perp},$$

which has dimension $m - \operatorname{rank}(F)$.

Theorem 8.2 (Projective rank reliability)**.** *The probability of retaining at least t independent target recovery equations is*

$$R_t(s) = \sum_{\substack{F \subseteq \mathrm{PG}(m-1,q) \\ \mathrm{rank}(F) \le m-t}} (1-s)^{|F|} s^{N-|F|}. \tag{38}$$

Writing $N_v = (q^v - 1)/(q-1)$, this equals

$$R_t(s) = \sum_{u=0}^{m-t} \begin{bmatrix} m \\ u \end{bmatrix}_q \sum_{v=0}^{u} \begin{bmatrix} u \\ v \end{bmatrix}_q (-1)^{u-v} q^{\binom{u-v}{2}} s^{N-N_v}. \tag{39}$$

Its endpoint terms are

$$R_t(s) = \begin{bmatrix} m \\ t \end{bmatrix}_q s^{M_t} + O(s^{M_t+1}) \qquad (s \to 0), \tag{40}$$

$$1 - R_t(s) = B_{m,m-t+1}(1-s)^{m-t+1} + O((1-s)^{m-t+2}) \qquad (s \to 1), \tag{41}$$

where

$$B_{m,r} = \begin{bmatrix} m \\ r \end{bmatrix}_q \frac{|\,\mathrm{GL}(r,q)|}{r!(q-1)^r}$$

is the number of unordered projective r-frames.

Proof. Recovery of t independent target combinations is possible exactly when $m - \mathrm{rank}(F) \ge t$, which gives (38). For a vector subspace $U \le \mathbb{F}_q^m$, let $\mathbb{P}(U)$ denote its projective points. Möbius inversion in the subspace lattice gives the total probability of failed sets spanning a fixed u-space U as

$$\sum_{V \le U} \mu(V,U) s^{N-N_{\dim V}}, \qquad \mu(V,U) = (-1)^{u-v} q^{\binom{u-v}{2}}, \qquad v = \dim V.$$

Here the total weight of all failed sets contained in $\mathbb{P}(V)$ is

$$\sum_{F \subseteq \mathbb{P}(V)} (1-s)^{|F|} s^{N-|F|} = s^{N-N_{\dim V}},$$

which accounts for the power of s before inversion. There are $\begin{bmatrix} m \\ u \end{bmatrix}_q$ choices for U and $\begin{bmatrix} u \\ v \end{bmatrix}_q$ subspaces $V \le U$ of dimension v. Summing over $u \le m-t$ proves (39).

For $s \to 0$, a smallest surviving set that permits rank-t recovery is the complement of the points in an $(m-t)$-dimensional vector subspace. It has size M_t, and there are $\begin{bmatrix} m \\ m-t \end{bmatrix}_q = \begin{bmatrix} m \\ t \end{bmatrix}_q$ such subspaces. For $s \to 1$, failure first occurs when the failed points span dimension $m-t+1$. A smallest such set is a projective frame of that size. Counting its span and then its unordered projective bases gives $B_{m,m-t+1}$. □

At the same helper length, the simplex presentation is characterized by its first recovery cost. This is the length-N, multiplicity-one specialization of Bonisoli's classification of linear equidistant codes [4]; the proof below records the precise zero-column and projective-multiplicity conventions used here.

Proposition 8.3 (Equality in the first projective bound)**.** *Let B be any full-row-rank $m \times N$ matrix over $\mathbb{F}_q$, with $N = (q^m-1)/(q-1)$. Then*

$$\min_{0 \ne a \in \mathbb{F}_q^m} \mathrm{wt}(aB) \le q^{m-1}.$$

Equality holds if and only if, up to column scaling and permutation, the columns of B represent every point of $\mathrm{PG}(m-1,q)$ exactly once.

Proof. For each nonzero column, the fraction of nonzero $a \in \mathbb{F}_q^m$ for which its entry in aB is nonzero is

$$\frac{q^{m-1}(q-1)}{q^m - 1}.$$

Averaging $\mathrm{wt}(aB)$ over $a \neq 0$ gives at most q^{m-1}, strictly less if B has a zero column. If the minimum attains this average, every nonzero linear combination has weight q^{m-1} and every projective hyperplane contains total column multiplicity $(q^{m-1}-1)/(q-1)$.

Let c be the vector of projective column multiplicities. The point–hyperplane incidence matrix is square, and its real Gram matrix has the form $(r-\lambda)I + \lambda J$ with $r > \lambda$. It is nonsingular, so the constant hyperplane-sum equations have the unique solution $c = \mathbf{1}$. Conversely, the simplex matrix has constant nonzero weight q^{m-1}. □

9 Verification boundary

The paper has a paper-owned Lean 4 companion built against a pinned Mathlib revision. It formalizes the construction of the associated nested code pair from abstract target and helper generator maps. In the notation of (1), the checked statements are

$$K_P \leq D_P, \qquad G_J(D_P) = W_P, \qquad G_J|_{D_P} : D_P \twoheadrightarrow W_P, \qquad \ker(G_J|_{D_P}) = K_P.$$

The corresponding declarations are

Paper statement	Lean declaration
$K_P \leq D_P$	`helper_ker_le_helperCodeForTargetSpace`
$G_J(D_P) = W_P$	`map_helperCodeForTargetSpace_eq_recoverableTargetMessageSpace`
surjectivity	`recoverableTargetMap_surjective`
restricted kernel	`mem_ker_recoverableTargetMap_iff`

The companion uses Lean 4.32.0-rc1 and the Mathlib commit

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

The axiom audit of all four reviewer-facing declarations reports only

$$\{\texttt{Classical.choice}, \texttt{Quot.sound}, \texttt{propext}\}.$$

No coding-theory result is introduced as a Lean axiom.

The formal coverage stops at the exact sequence. In particular, Proposition 3.2, Theorems 3.3, 4.1, 4.3, 4.4, and 4.5, as well as Proposition 4.2, the strictness construction in Proposition 4.6 and the later consequences and examples, are human-proved and are not established by the paper-local Lean companion. The claim manifest records them as absent rather than treating the exact-sequence formalization as evidence for them.

The reliability polynomials in Proposition 7.1 follow from the displayed union-size table by inclusion–exclusion. No exhaustive computation is a premise of that proposition or of any theorem in the main proof chain. Machine-readable examples and replay checks may therefore be used to audit arithmetic without changing the logical dependence of the paper.

10 Conclusion

For a fixed target set, shortening and puncturing the inner dual onto the helper coordinates give the associated nested code pair. Its relative generalized Hamming weights give the exact minimum helper unions for recovered subspaces of each dimension. The full normalized recovery equations retain the coefficient and overlap data that those minima discard.

The dual nested pair describes the same recovery problem from the failure side: its relative weights are the minimum numbers of helper failures that leave prescribed dimensions of target ambiguity. Thus helper cost, erasure ambiguity, and rank reliability are different summaries of the recovered dimension as the surviving helper set varies.

The finite concatenation theorem minimizes prescribed-coset support costs over the complete outer functional dual. Its zero-functional sector has cost

$$M_t(D_P, K_P) + d(I^{\perp}).$$

After the outer-distance gate, every nonzero-functional sector lies above the chosen radius, so this additive expression gives the exact confinement criterion. In sufficiently long concatenations whose outer dual distance grows, every bounded equation below the corresponding threshold remains in its inner block and zero-extension copies its coefficients and exact support. This yields positive-density realizations and eventually transfers any consequence that depends only on the copied minimal recovery sets, including bounded repair reliability and service-rate regions.

At one target coordinate, the same finite formula becomes the scalar target-constrained coset-weight formula. The Singer-cycle construction shows that this specialization can certify transfer when ordinary outer support distance cannot.

Under repeated concatenation, the labelled ordinary prescribed-coset support functions compose by exact min-sum substitution. Helper-restriction costs and target images give the target-normalized recursion, while the inner-dual distance obeys a compatible formula. Thus the finite nonconfinement cost can be evaluated through a tower while keeping its zero and nonzero functional sectors distinct. The persistent scalar $\rho_T(I) + d(I^{\perp})$ does not determine the next-level cost.

The same labelled recursion is executable. ERGO-Comp evaluates it by an exact min–sum dynamic program, retains block labels and coefficient-level witnesses, and couples the resulting recovery choices to an exact capacity-aware scheduler. Quotient compilation and positive gradings can replace a large ambient box by its reachable state set without weakening the answer. Direct and equivalently preprocessed CP-SAT comparisons separate the benefit of this algebraic compilation from the benefit of the specialized state engine.

The two separations locate the remaining information. Equal relative weights do not determine the stochastic law because they do not retain the intersection pattern of all recovery sets. Even the associated nested pair does not determine the additive threshold: different ambient inner-dual realizations can change $d(I^{\perp})$. This is distinct from changing the generator-row basis of one inner code. The projective-simplex family exhibits both levels explicitly: its relative weights give the minimum recovery costs, while the projective rank distribution gives the complete dimension-by-dimension reliability law.

Two extensions require different objects. Disjoint simultaneous service is an integral packing problem, rather than a minimum-union or fractional service-rate problem. Bandwidth-aware repair for array codes allows each helper to transmit a subspace of functionals, so its cost is not Hamming support. Neither extension follows by changing terminology in the present theorems; each requires a confinement invariant adapted to its cost model.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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